

ISRO Hackathon: Development of Surface AQI & Identification of HCHO Hotspots over India using Satellite Data

1. Executive Summary & Overview

Air pollution is a critical global health hazard, responsible for millions of excess deaths annually. Communicating ambient air pollution levels (comprising pollutants such as CO, NO₂, O₃, SO₂, and PM_{2.5}) is typically achieved using the **Air Quality Index (AQI)**.

The Core Problem

- Ground Monitor Scarcity:** Most of the global population lives further than 100 km away from ground air quality monitors. This leaves vast geographical regions devoid of real-time air quality information.
- Spatio-Temporal Gaps:** Traditional ground monitoring cannot provide contiguous spatial maps of daily air pollution. Leveraging satellite observations is critical to bridge this gap.
- Volatile Organic Compounds (VOCs) & Biomass Burning:** Formaldehyde (HCHO) is a key indicator of VOC emissions, playing a vital role in ozone formation and atmospheric chemistry. During biomass burning events (e.g., agricultural residue burning and forest fires), massive amounts of VOCs are released. Spatio-temporal analysis of HCHO hotspots is required to understand these emissions, their sources, and their atmospheric transport.

Regulatory and National Alignments

- National Clean Air Programme (NCAP):** India's long-term, time-bound strategy to reduce air pollution by **20–40%** in non-attainment cities.
- Sustainable Development Goals (SDG):** Directly aligns with **SDG Target 11.6**, which calls for reducing the adverse environmental impacts of cities.
- Chintan Shivir 2.0:** Targets urban climate resilience, air quality monitoring, mitigation, and sustainable development.
- Relevance:** This problem statement is directly relevant to **ISRO** and Indian national priorities.

2. Core Objectives

Objective 1: Development of Surface Air Quality Index (AQI)

- Goal:** Create a daily surface AQI estimation system using satellite observations.
- Output:** Generate continuous spatial maps of AQI over the entire Indian landmass.
- Methodology:** Use deep learning to integrate columnar satellite data with ground observations and meteorological datasets to deduce ground-level PM_{2.5} and other key AQI inputs.

Objective 2: High-Resolution Mapping of HCHO Hotspots

- Goal:** Identify and analyze spatio-temporal hotspots of Formaldehyde (HCHO) over India.
- Output:** High-resolution spatial maps during biomass burning seasons.
- Methodology:** Map the distribution, detect sources, correlate with fire activity, and model atmospheric transport.

Objective 3: Source Identification & Transport Analysis

- **Goal:** Delineate major source regions contributing to pollution episodes.
- **Key Areas:** Focus on the **Indo-Gangetic Plain (IGP)** and active forest fire zones (e.g., Northeast India, Central India).
- **Transport Influence:** Evaluate how wind currents and meteorological reanalysis transport pollutants away from source zones.

3. System Flow Architecture

```
graph TD
  %% Objective 1 Flow
  subgraph Objective_1 ["Objective 1: Surface AQI Model Pipeline"]
    A1[INSAT-3D AOD Satellite Data] --> D1[Database Pre-processing & Feature Alignment]
    A2[CPCB Ground Station Data] --> D1
    A3[Meteorological Reanalysis ERA5/IMDAA/MERRA-2] --> D1
    D1 --> E1[Deep Learning Model CNN-LSTM / LSTM / CNN]
    E1 --> F1[Validate Predictions vs CPCB Ground Truth]
    F1 --> G1[Generate Daily Spatial Maps of Surface AQI over India]
  end

  %% Objective 2 Flow
  subgraph Objective_2 ["Objective 2: HCHO Hotspots & Transport Pipeline"]
    B1[Sentinel-5P HCHO Column Data] --> D2[Acquisition & Spatio-temporal Grid Alignment]
    B2[Fire Count Datasets MODIS/VIIIRS] --> D2
    B3[Meteorological Wind & Transport Data] --> D2
    D2 --> E2[Statistical Thresholding / Clustering Models]
    E2 --> F2[HCHO Hotspots & Source Region Identification]
    F2 --> G2[Fire-HCHO Correlation & Wind Transport Analysis]
  end
```

4. Required Datasets & Official Access Links

The table below outlines the core datasets required for this project, along with official repositories and access links.

Dataset Name	Description / Variables	Sensor/Satellite Platform	Official Download & Access Links
INSAT-3D AOD	Aerosol Optical Depth (AOD) data product used to deduce surface PM2.5.	INSAT-3D Imager	MOSDAC INSAT-3D Data Products
Sentinel-5P Columns	Columnar concentrations of NO2, SO2, CO, O3, and HCHO.	TROPOMI (TROPOspheric Monitoring Instrument)	Google Earth Engine Catalog DLR Geoservice S5P L3 Offline Data
CPCB Ground Data	Ground-based measured concentrations of PM2.5, PM10, NO2, etc.	Continuous Ambient Air Quality Monitoring (CAAQMS)	CPCB CAAQM Dashboard & Repository
Fire Count Data	Spatial distribution, Fire Radiative Power (FRP), and fire occurrence counts.	MODIS (Terra/Aqua) & VIIRS (Suomi NPP/NOAA-20)	NASA FIRMS Active Fire Data
Meteorological Reanalysis	Reanalysis parameters including wind profiles, temperature, boundary layer height, etc.	ERA5, IMDAA, or MERRA-2	Copernicus Climate Data Store (CDS ERA5) NCMRWF IMDAA Reanalysis NASA GES DISC MERRA-2 Project

5. Insight Recorder: Required Variables & Features

Below are the comprehensive schemas of required parameters to extract, clean, and integrate from each dataset. These represent the specific variables and insights necessary for successful model building.

5.1 INSAT-3D Satellite Data

Used primarily for deriving surface aerosol and particulate indices.

Variable / Parameter	Priority	Notes / Description
AOD (Aerosol Optical Depth)	High	Core predictive input for ground PM2.5 retrieval.
Latitude & Longitude	High	Geospatial grid coordinate referencing.
Cloud Mask	High	Filter out cloud-contaminated pixels.
QA Flag	High	Quality Assurance flags to filter low-confidence retrievals.
Solar Zenith Angle	High	Geometric correction indicator.
Satellite Zenith Angle	High	Viewing geometry indicator.
Relative Azimuth Angle	High	Angular relationship between sun and sensor path.
Observation Time	High	Exact timestamp for temporal matching.
Surface Reflectance	High	Background land surface reflectance correction.

5.2 Sentinel-5P TROPOMI Satellite Data

Provides column densities of trace atmospheric gases.

Variable / Parameter	Priority	Notes / Description
HCHO	High	Formaldehyde column density (mol/m²). Key VOC indicator.
NO2	High	Nitrogen Dioxide column density. Traces combustion/traffic.
SO2	High	Sulfur Dioxide column density. Industrial and volcanic marker.
CO	High	Carbon Monoxide column density. Biomass/combustion marker.
O3	High	Ozone column density. Secondary atmospheric pollutant.
CH4	High	Methane column density. Greenhouse gas and VOC component.
Aerosol Index	High	Identifies UV-absorbing aerosols (dust, smoke).

Cloud Fraction	High	Proportion of the pixel obscured by cloud cover.
QA Value	High	Quality value indicating retrieval reliability (>0.5 or >0.75 recommended).
Air Mass Factor	High	Radiative transfer calculation parameter for vertical columns.

5.3 CPCB (Central Pollution Control Board) Ground Data

Ground truth datasets used to train models and validate satellite-derived estimations.

Variable / Parameter	Priority	Notes / Description
PM2.5 & PM10	High	Fine and coarse particulate matters ($\mu\text{g}/\text{m}^3$).
NO2, SO2, CO, O3, NH3	High	Core Criteria Gaseous Pollutants.
Benzene, Toluene, Xylene	High	Specific organic aromatic compounds (VOC indicators).
AQI (Air Quality Index)	High	Officially calculated Ground AQI.
Temperature, Humidity, Pressure	High	Surface meteorological measurements at ground stations.
Wind Speed & Wind Direction	High	Surface wind variables at ground stations.
Rainfall	High	Precipitation indicator (wet deposition parameter).

5.4 Fire Activity Data (MODIS / VIIRS)

Identifies location, intensity, and timing of open agricultural burning and forest fires.

Variable / Parameter	Priority	Notes / Description
Fire Count	High	Frequency of detected thermal anomalies in grid cells.
FRP (Fire Radiative Power)	High	Radiant energy output of fire (indicates fire intensity/size, MW).
Confidence	High	Detection confidence percentage (0 - 100%).
Brightness Temperature	High	Channel 21/22 and Channel 31 temperatures.
Latitude & Longitude	High	Spatial coordinates of active fire pixels.
Date & Time	High	Time stamp of observation (UTC/IST).
Day/Night	High	Flag indicating if fire was detected during day or night orbit.

5.5 Meteorological Reanalysis Data (ERA5 / IMDAA / MERRA-2)

Used for atmospheric modeling, boundary layer estimation, and covariate modeling.

Variable / Parameter	Priority	Notes / Description
2m Temperature	High	Air temperature at 2 meters above ground.
Skin Temperature	High	Temperature of the physical land surface.
Surface Pressure	High	Local atmospheric pressure at surface height.
MSLP (Mean Sea Level Pressure)	High	Normalized pressure at sea level.
10m U Wind & 10m V Wind	High	Zonal (East-West) and Meridional (North-South) wind components.
Wind Speed & Wind Direction	High	Derived absolute wind velocity and transport angle.
Relative & Specific Humidity	High	Atmospheric moisture indicators.
Dew Point	High	Dew point temperature at 2 meters.
Precipitation	High	Total accumulated rainfall/snowfall over grid area.
Cloud Cover	High	Fraction of grid covered by clouds.
PBL Height	High	Planetary Boundary Layer height (critical for surface dilution volume).
Solar Radiation	High	Incoming shortwave radiation (drives photochemical reactions).
Soil Moisture	High	Volumetric soil water content (affects dust emission potential).

5.6 Auxiliary Land Surface Data

Provides static and dynamic surface features that influence pollution retention and dispersion.

Variable / Parameter	Priority	Notes / Description
NDVI & EVI	High	Vegetation indices (identifies crop health and biomass load).
Land Surface Temperature (LST)	High	Radiative surface temperature.
Burned Area	High	Total post-fire mapped scar area.
Land Cover	High	Land Use Land Cover (LULC) class (forest, crop, urban, barren).
Elevation, Slope, Aspect	High	Topographical parameters (critical for terrain-induced trapping).
Night-time Lights	High	Proxy for human settlement and continuous night energy usage.
Population Density	High	Anthropogenic exposure mapping.
Road & Industrial Density	High	Spatial proxies for transportation and point-source emissions.

5.7 Derived / Feature Engineered Variables

Custom indicators to be calculated during feature engineering to boost model performance.

Variable / Parameter	Priority	Formula / Description
Daily, Weekly, Monthly Mean	High	Rolling averages of gaseous pollutants.
Moving Average	High	Short and long-term lag variables.
Prev AQI, PM2.5, HCHO	High	Autoregressive lag features (e.g., t-1, t-2 days).
Distance to Fire	High	Distance from target grid center to closest active fire pixels.
Fire Count Buffers	High	Total fire counts within radius zones: 5km, 10km, 25km, 50km.
Mean FRP	High	Average Fire Radiative Power in neighborhood zones.
Wind Transport Vector	High	Vector product of Wind Speed, Direction, and upstream Fire Count.
Season & Day of Year	High	Temporal cyclic sine/cosine variables mapping yearly cycles.
Crop Burning Indicator	High	Boolean indicator combining LULC (cropland) and active Fire counts.

6. Implementation Steps

Objective 1 Steps: Surface AQI Model Development

- Grid Standardization:** Set up a uniform spatial grid over the Indian landmass (e.g., 0.1 degrees x 0.1 degrees or 0.05 degrees x 0.05 degrees).
- Satellite AOD Processing:** Extract, quality-filter (using QA flags), and interpolate INSAT-3D AOD observations onto the grid.
- Meteorological Alignment:** Downscale and interpolate ERA5/IMDAA/MERRA-2 parameters (especially PBL height, wind, humidity) to the target grid.
- CPCB Ground Station Collocation:** Pair ground monitor measurements with corresponding satellite grid cells and temporal windows (hourly/daily).
- Model Architecture Design:** Develop a hybrid deep learning model (e.g., **CNN-LSTM**). The CNN extracts spatial features from satellite datasets, while the LSTM handles temporal patterns and lag effects (e.g., yesterday's pollution, weekly cycles).
- Validation and Mapping:** Validate the model using a spatial holdout approach (testing on stations the model has never seen). Compute daily predictions for all grid cells to generate spatial maps of AQI over India.

Objective 2 Steps: HCHO Hotspots and Fire Correlation

- S5P HCHO Processing:** Retrieve TROPOMI Sentinel-5P HCHO vertical column densities. Filter using QA values (>0.5).
- Biomass Fire Extraction:** Retrieve active fire coordinates, FRP, and times from MODIS/VIIRS.

3. **Spatio-Temporal Clustering:** Deploy hotspot detection algorithms (e.g., Getis-Ord G_i^* statistics, DBSCAN, or Kernel Density Estimation) to isolate statistically significant HCHO anomalies.
4. **Correlation Studies:** Calculate the mathematical correlation between fire intensity (FRP x Fire Count) and downwind HCHO enhancements during crop residue burning seasons (e.g., October–November in Northwest India, March–April in Central/NE India).
5. **Transport Modeling:** Use 10m/boundary-layer wind vectors to trace the downwind plume dispersion and estimate pollutant transport trajectories.

7. Suggested Technical Stack

- **Programming Languages:** Python, JavaScript (for Google Earth Engine scriptings), MATLAB (alternative processing).
- **Cloud Platforms:** Google Earth Engine (GEE) API for fast spatial queries, Sentinel-5P cataloging, and MODIS ingestion.
- **Deep Learning Frameworks:** PyTorch / TensorFlow (implementing CNN, LSTM, and CNN-LSTM models).
- **Data Processing Libraries:** Pandas, GeoPandas, Xarray, Rasterio, NumPy, OpenPyXL (for spreadsheet integration).
- **Visualization Tools:** Matplotlib, Seaborn, Folium, QGIS, or GEE Explorer.

8. Evaluation Metrics

Objective 1 Evaluation: Surface AQI Predictions

To evaluate model performance against CPCB ground truth stations, compute the following metrics: * **Root Mean Squared Error (RMSE):** Measures the magnitude of estimation errors. $RMSE = \sqrt{\text{Sum}((Y_i - Y_{pred_i})^2) / N}$ * **Pearson Correlation Coefficient (R / R^2):** Measures spatial and temporal alignment strength. * **Mean Absolute Error (MAE):** Average absolute prediction error. $MAE = \text{Sum}(|Y_i - Y_{pred_i}|) / N$

Objective 2 Evaluation: HCHO Hotspots & Transport

- **Hotspot Detection Accuracy:** Intersection over Union (IoU) of predicted hotspots vs true high-concentration zones.
- **Multi-Source Dataset Integration Quality:** Verification of continuous overlap across datasets (S5P, MODIS, ERA5).
- **Scientific Interpretation & Plausibility:** Alignment of detected transport paths with known meteorological regimes.
- **Visualization Quality:** Clarity, resolution, and readability of the spatial-temporal maps over the Indian region.
- **Methodological Innovation:** Value added by novel neural architectures or custom physical features (e.g., wind transport vectors).